A Lunar Ground Truth of Microbes that are integral to sustaining Biological Life Support Systems

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A lunar outpost such as Gateway or a Mars transit vehicle will use a semi-closed to closed loop biological life support system (BLSS) adapted for microgravity and high levels of radiation. A moon habitat will likely exist under partial gravity conditions and sustained levels of high radiation as a semi-closed loop system able to get resources from Earth via Gateway. Additionally, the moon will act as a closed-loop testbed for Mars habitat operations and will employ various elements of in situ resource utilization (ISRU). Perhaps the longest running BLSS collective study to date is that of the European MELiSSA (Micro Ecological Life Support System Alternative) project, a circular life support system, established to gain knowledge on regenerative systems aimed at the highest degree of autonomy to produce food, water and oxygen from mission wastes. This setup has evaluated the use of discrete microbial compartments and a higher plant compartment to carry out the necessary life support functions to support crew. Such compartments utilize thermophilic anaerobes to break down human and inedible plant wastes, photoheterotrophic bacteria that can further metabolize volatile fatty acids, nitrifying bacteria that can convert ammonium to plant and microalgae available nitrates, and photoautotrophic bacteria and higher plants which will convert carbon dioxide to oxygen, purify water, and provide food for human consumption. This being the most advanced BLSS model example to date -aside from the Yuegong-1, Chinese Lunar Palace- with the most defined compartment composition, and the assumption that other BLSSs will build upon such models, it becomes relevant to study the effects of the lunar environment upon the microbes that are integral to a BLSS. In ramping up to advanced stage BLSS systems, relatively simple experiments can be conducted on Artemis missions using continuous culturing and sampling of the model microbes, subsequently subjected to DNA sequencing for mutational analysis monitoring and chemical analysis to assess the sustained ability to carry out their hallmark biochemical processes efficiently. A suite of microbes should be assessed, prioritizing those with the duel capacity to be utilized in an BLSS and the ability to biochemically facilitate ISRU goals, such as the ability to transform the biogeochemistry of Moon or Martian regolith into materials that can support crop growth or extract elements of industrial significance such as aluminum or iron.